SVP Partner Engagement & Analytics Dashboard

DASS Spring 2026 — Team 49 | Release 1 Presentation

Client: Raja Prasada Rao Poladi
Course: Design and Analysis of Software Systems

Presentation By –

- Prashant Vinod (2024101116)
- Athmeeya Kashyap (2024113015)
- Navneet Kumar Gupta (2024111030)
- Sambhavi Singh (2024117010)
- Somisetty Yogansh (2024101006)

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

IIIT Hyderabad

SVP INDIA

What is SVP?

Social Venture Partners (SVP) is a global network of engaged philanthropists, connecting skilled professionals with non-profits to create structured impact. Each country has its own independent operations.

Previously, the Hyderabad chapter managed these critical interactions manually, leading to severe operational bottlenecks:

- No centralized system for tracking engagements
- Scattered attendance records via spreadsheets/WhatsApp
- Double-bookings on recurring mentorship sessions
- No visibility for leadership to make data-driven decisions

SVP INDIA

Problem Statement

The **SVP Hyderabad chapter** was managing partner engagement manually.

Current Workflow

- Meetings tracked through **spreadsheets and emails**
- Attendance tracked using **paper or WhatsApp messages**
- Recurring meetings scheduled **manually every time**

Challenges

- No centralized system for tracking meetings
- Difficult to monitor partner participation
- Recurring meetings prone to scheduling errors
- Hard for leadership to measure reliable engagement

Impact

- Missed follow-ups
- Poor visibility into partner activity
- Manual reporting and analytics

Client Expectations

- Mr. Raja Poladi, Head of CDiT-H, provided an informal requirements document describing how different entities in the system should interact and how each view of the platform should function. Based on this guidance, the client expected us to translate these workflows into a **structured and intuitive web application**.

- A key expectation was to ensure the platform is easy to use and significantly **reduces the time** and **effort currently spent by SVP Team** in managing partner engagement through manual processes. The system also needed to be designed with good software engineering practices, ensuring it is **modular**, **maintainable**, and **scalable** for future releases.

Our Solution

Our SDLC is a **prototyping model** and we have developed a centralized full-stack web platform called:

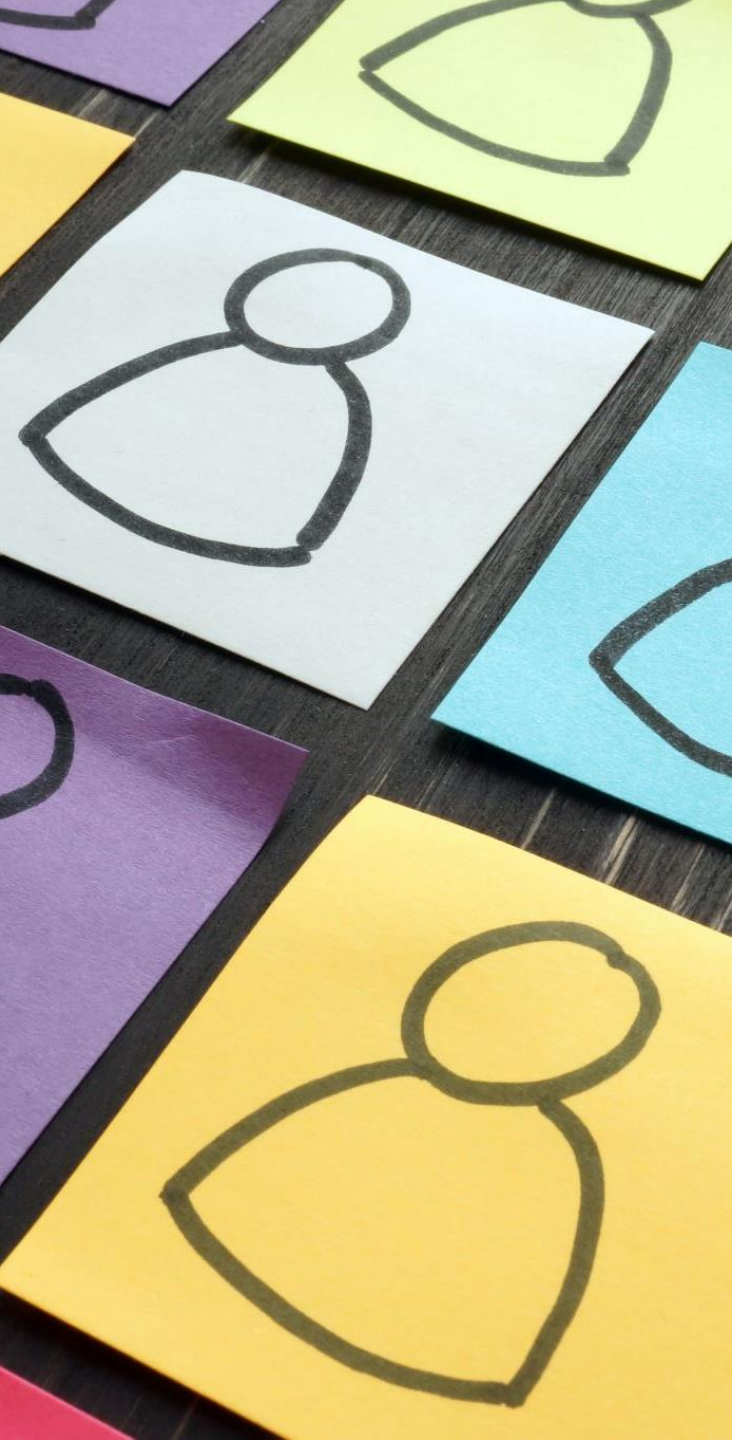
SVP Partner Engagement Dashboard

Objectives

- Replace manual spreadsheets with a structured platform
- Track all partner engagement in a **single system**
- Provide **analytics and reporting capabilities**
- Simplify meeting scheduling and attendance tracking

Core Capabilities

- Manage **Partners, Investees, and Groups**
- Schedule **appointments and recurring meetings**
- Track **attendance per partner**
- Visualize **engagement analytics**



Partner & Investee Management

The system allows administrators to manage partners and organizations efficiently.

Partner Management

- Create and update partner profiles
- Search partners by **name or email**
- Sort partners by **start date, email, or name**
- View detailed **appointment history and attendance**

Additional Feature

- **Primary ↔ Secondary partner hierarchy**

Investee Management

- Maintain records of all investee organizations
- View all meetings associated with a specific investee

Group Management

Groups represent teams of partners collaborating with an investee.

Features

- Create groups linked to an investee
- Define group types: Mentorship, Board, Skills Training, Fund Committee, Operations, Outreach
- Add or remove partners from groups
- Expandable view to see group members

Benefit

- Enables quick scheduling by attaching **all group partners to meetings at once**.

Appointment Scheduling

The system includes a **calendar-based meeting scheduler**.

Calendar Interface

1) Monthly calendar view
2) List view for browsing appointments

Appointment Creation
Admins can schedule meetings with:

1) Meeting type 2) Date & time 3) Linked investee
4) Assigned partners

Meeting Status

1) Pending 2) Completed 3) Cancelled

Attendance Tracking

Mark each partner as **Present ✓ or Absent ✕**

Recurring Appointments (Core Feature)

Many SVP meetings repeat **weekly or monthly**.

Scheduling these manually leads to errors and extra work.

Our Solution

We implemented **Recurring Appointment Templates**.

Implementation

- Uses **RRule standard**
- Supports:

1) Weekly meetings 2) Biweekly meetings 3) Monthly meetings

Example

Mentorship meeting
Every Monday
10:00–11:00 AM

Analytics Dashboard

The system includes a dashboard that provides **engagement insights**.

Key Metrics

- Total meetings conducted
- Number of active partners
- Total volunteer hours
- Active investee organizations

Visualizations

- **Attendance by Partner** – bar chart
- **Monthly Engagement Trends** – time series chart
- **Meeting Categories** – breakdown by meeting type
- **Investee Analytics** – engagement per organization

These analytics help leadership **make data-driven decisions.**

Progress So Far (Release 1)

- For Release 1, we have completed the major phases of the R1 SDLC including **requirements elicitation, requirement specification, system architecture design, detailed design, and implementation**. The core features of the SVP Partner Engagement Dashboard have been successfully developed.

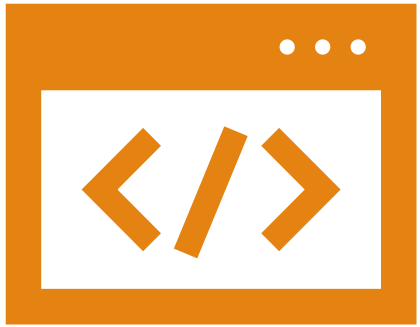
- At this stage, the project has progressed up to the **unit testing phase**, where individual components and modules of the system are being verified to ensure correct functionality. The system is now stable at the module level and ready to proceed to **integration testing and further validation with the client**.

System Architecture

The application follows a **modern full-stack architecture**.

Frontend

- React + TypeScript
- Tailwind CSS for UI styling
- Chart.js for analytics visualization



Backend

- Node.js with Express REST APIs
- Controller → Service → Repository architecture
- Prisma ORM for database operations

Database

- PostgreSQL relational database
- Multiple relational tables for partners, groups, appointments, and recurring meetings

Additional Components

- JWT authentication
- Cron jobs for recurring appointment generation
- Swagger API documentation

Authentication & Security

Security is implemented using **JWT-based authentication**.

Login System

- Email and password authentication
- Tokens generated using **JSON Web Tokens**

Password Security

- Passwords stored using **bcrypt hashing**

API Protection

- Middleware verifies tokens before API access
- Protected frontend routes

Additional Measures

- Controlled admin access
- Consistent error handling
- Protection against email enumeration

Challenges Faced till Release 1

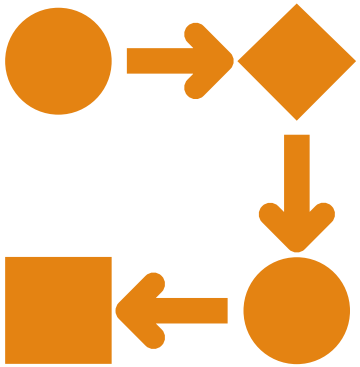
- Coordination and communication challenges within the team and with the client during development
- Maintaining **consistent understanding of requirements across the team** during development
- Difficulty designing the **recurring appointment logic**, requiring multiple discussions with the client
- Converting the client's **informal requirement document into structured system requirements**
- Learning and evaluating the required **technology stack** during the development phase
- Managing **version control and code integration** when multiple team members worked on different modules

Demo Walkthrough

Future Work (Release 2)

In the next phase, we will focus on completing the system delivery and preparing for the next release.

- Perform **rigorous integration testing** to ensure system stability
- Conduct **System Acceptance Testing** and **Client Acceptance Testing**
- **Deploy the application** and hand over the system for real usage
- **Observe system usage** and collect feedback from the users
- Gather **new requirements (which also include feedbacks from Release 1) for Release 2** and begin a **new SDLC cycle**



THANK YOU

Questions?